$pp \to e^+e^-$ (DY, $m_{ee} > 60$ GeV): 1000000 events, 13 TeV, NNPDF4.0 NLO, MC@NLO-Δ source

Shower: Pythia 8.318 from the first-emission endpoints, without MPI, hadronisation, primordial k_T, QED or top/Higgs decays; 50 split jobs

Totals: baseline 1933 pb (+0.00%) S veto only 1906 pb (-1.36%) negative-H removal only 1960 pb (+1.44%) MAcNLOPS 1934 pb (+0.08%)

Ratio bands: correlated statistical uncertainty (per event at the first emission, batch means over the split shower jobs after the shower), where the baseline is known to 25%.

first emission (LHE)

after the shower

baseline (no veto, no removal)
S veto only
negative-H removal only
MAcNLOPS (veto and removal)

$d\sigma/d\log_{10} p_T$ [pb]

ratio to baseline

$p_T(e^+e^-)$ [GeV]

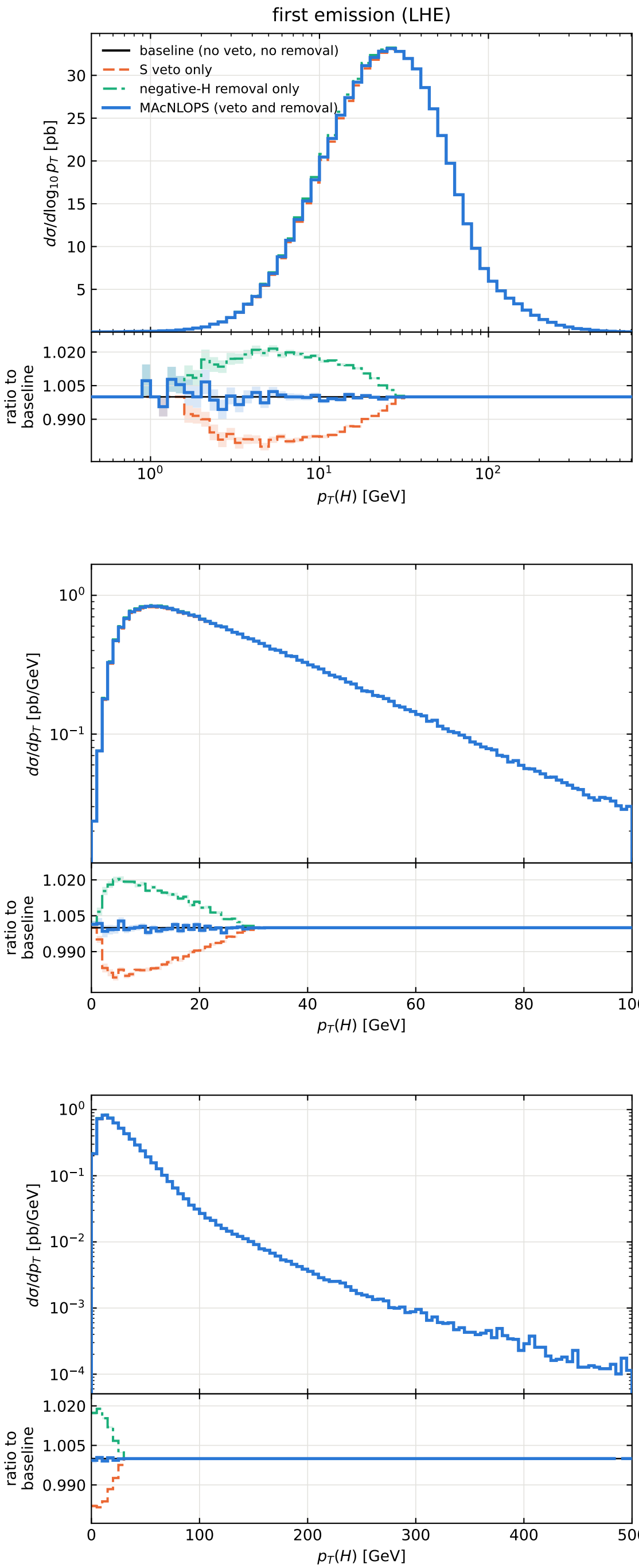
$d\sigma/dp_T$ [pb/GeV]

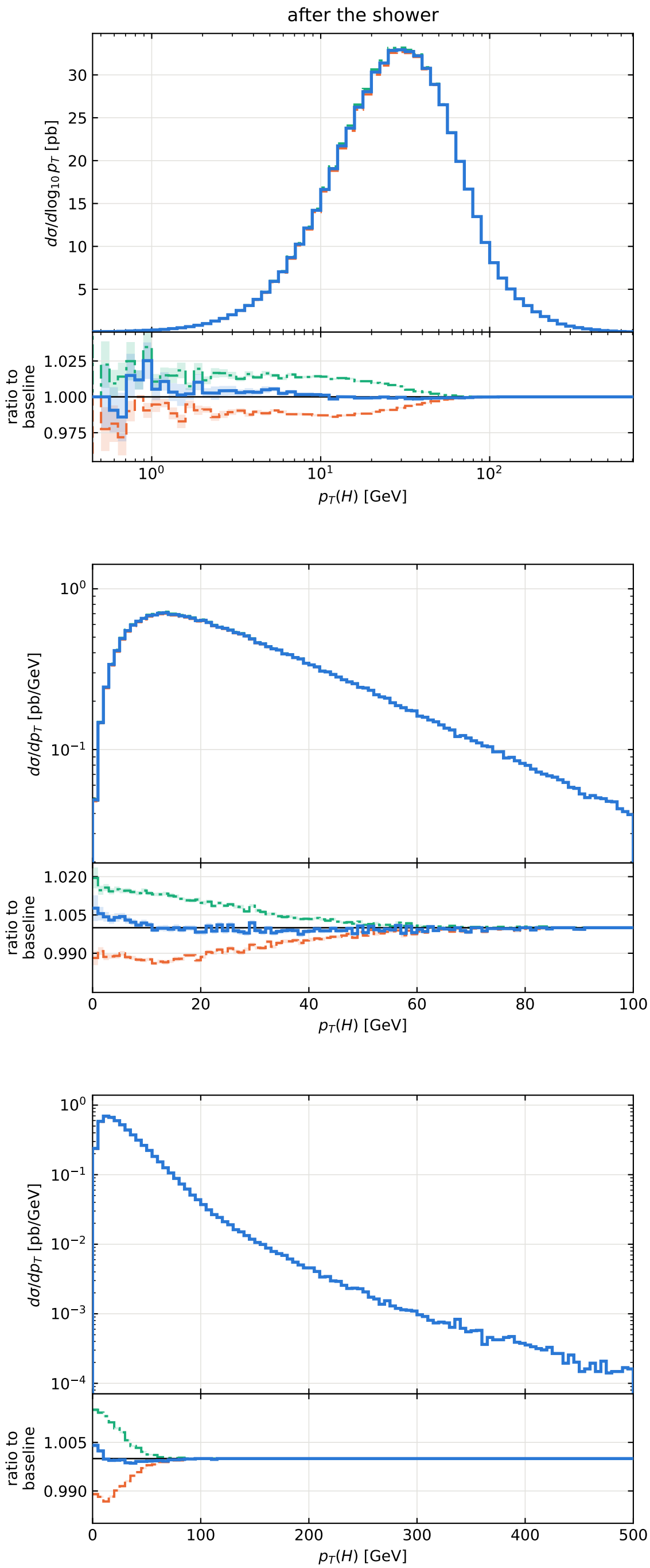
gg→H (HEFT): 1000000 events, 13 TeV, NNPDF4.0 NLO, MC@NLO-Δ source

Shower: Pythia 8.318 from the first-emission endpoints, without MPI, hadronisation, primordial k_T, QED or top/Higgs decays; 50 split jobs

Totals: baseline 30.77 pb (+0.00%) S veto only 30.54 pb (-0.74%) negative-H removal only 30.99 pb (+0.72%) MAcNLOPS 30.77 pb (-0.01%)

Ratio bands: correlated statistical uncertainty (per event at the first emission, batch means over the split shower jobs after the shower), where the baseline is known to 25%.





$pp \to t\bar{t}$: 1000000 events, 13 TeV, NNPDF4.0 NLO, MC@NLO-Δ source

Shower: Pythia 8.318 from the first-emission endpoints, without MPI, hadronisation, primordial k_T, QED or top/Higgs decays; 50 split jobs

Totals: baseline 637.5 pb (+0.00%) S veto only 559.3 pb (-12.28%) negative-H removal only 711.3 pb (+11.57%) MAcNLOPS 633 pb (-0.70%)

Ratio bands: correlated statistical uncertainty (per event at the first emission, batch means over the split shower jobs after the shower), where the baseline is known to 25%.

